

Trivariate frequency analysis using copulas emphasizing the importance of the duration of hydrographs for hydraulic works

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ABSTRACT

The use of copula functions in hydrology has increased significantly, as they allow for more precise hydrograph estimation by incorporating the joint behavior of key variables. However, modeling runoff events with three variables – flow, volume, and duration – presents challenges, especially due to the complexity in defining and discretizing the duration variable. This study evaluates the role of duration and proposes a methodology to estimate trivariate design hydrographs, giving priority to duration alongside flow and volume. A case study is presented for the La Angostura dam in Mexico, using a hybrid peak selection method combining the threshold and annual maxima approaches. Marginal distributions are estimated using maximum likelihood with genetic algorithms for single and mixed population models. Symmetrical and asymmetrical Archimedean trivariate copulas are applied, with the Clayton copula identified as the most appropriate model. Joint return periods are computed using the OR and Kendall's definitions. The proposed system allows any user-defined hydrograph shape to be applied to the estimated flow, volume, and duration values, ensuring flexibility in design representation. The results show that prioritizing duration enables the estimation of reliable hydrographs: short durations yield higher peaks and volumes, while longer ones lead to attenuated responses, improving the consistency of hydrological design.

Key words: asymmetric copulas, flood frequency analysis, hydrograph duration, joint return period, multivariate frequency analysis, symmetric trivariate copulas

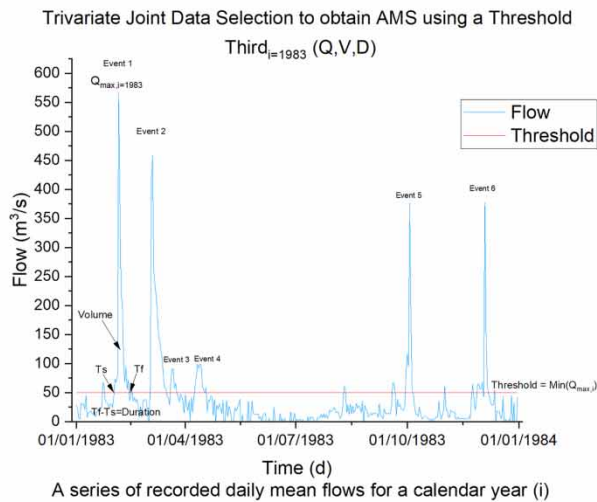
HIGHLIGHTS

- A hybrid data selection method prioritizing duration is proposed.
- Duration is the key variable in hydrograph estimation.
- Asymmetric and symmetric copulas model the relationships among flow, volume, and duration.
- Marginal probabilities are estimated to identify the best representation for the analysis.
- Joint return periods ensure an accurate representation of each probabilistic aspect.

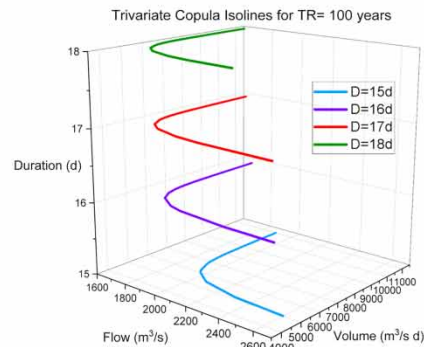
GRAPHICAL ABSTRACT

Trivariate Frequency Analysis Using Copulas Emphasizing the Importance of the Duration of Hydrographs for Hydraulic Works

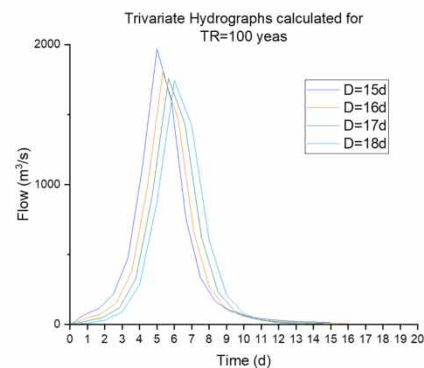
A trivariate joint hybrid method is presented, combining AMS and POT techniques for data selection to prioritize the duration variable. Annual maximum flow values are chosen, with the lowest serving as the threshold to define the start and end points of flood events. This approach eliminates personal judgment, ensuring a more objective selection of the Q, V, and D triplets and enhancing



Prioritizing the duration variable in hydrograph estimation improves the accuracy of runoff representation by considering a key hydrological factor often overlooked by traditional methods. This approach ensures all influencing factors are accounted for, eliminates personal judgment bias, and enhances variable correlation, resulting in more reliable hydrographs that better capture the temporal dynamics of runoff events.



Symmetrical and asymmetrical Archimedean family trivariate copulas were applied, identifying the Clayton copula as the most suitable model for this case. Primary joint return period OR and secondary Kendall's return period were calculated and compared to create hydrographs using the selected copula and applying a Pert function shape.



INTRODUCTION

Frequency analysis is commonly used in hydrology to estimate and analyze events associated with return periods, particularly in the design and assessment of hydraulic structures, flood estimation, risk analysis, and water resources management. Typically, flow is the primary variable considered, but this approach oversimplifies real events by failing to capture their multivariate nature. To address this limitation, multivariate frequency analysis, which incorporates multiple interrelated hydrological variables such as peak flow, volume, and duration, offers more comprehensive and reliable results.

Copulas, introduced by Sklar in 1979, are multivariate cumulative probability functions that estimate joint probabilities between variables by separating marginal distributions from their joint distribution. They provide a flexible and robust framework for modeling complex dependencies and have become essential in hydrology for capturing the interdependence between variables. As such, copula-based multivariate frequency analyses are better suited to representing the random variability inherent in natural runoff processes, which are typically defined by multiple correlated variables such as peak discharge, runoff volume, and flood duration (Goel *et al.* 1998; Salvadori 2004; Shiau *et al.* 2006).

Studies by Goel *et al.* (1998), Salvadori & De Michele (2004), and Shiau *et al.* (2006) have applied bivariate frequency analysis with copulas, typically using peak flow and volume for flood estimation. While this approach improves reliability compared with univariate models, it restricts the capacity to incorporate event duration explicitly – a variable closely linked to the hydrograph's shape and storage implications. Duration plays a critical role in both volume and peak discharge estimation and should not be treated as a secondary or fixed parameter.

However, defining event duration from observed data presents challenges. It is often selected based on the analyst's visual assessment of the start and end of a flood event, which introduces subjectivity and limits reproducibility (Zhang & Singh 2007; Ganguli & Reddy 2013). Several methods – statistical, heuristic, or based on watershed characteristics such as time of concentration – have been proposed to define duration, but the inherent variability of hydrological events makes consistent application difficult. These uncertainties can reduce the accuracy of hydrological models and affect decision-making in water management and infrastructure planning.

A multivariate frequency analysis that explicitly includes duration – alongside peak discharge and volume – offers a more objective framework by treating duration as a stochastic variable within the joint probability structure (Grimaldi & Serinaldi 2006). However, flow, volume, and duration are intrinsically interdependent, and their relationship can only be approximated using flexible copula structures (Serinaldi & Kilsby 2013). This justifies the use of trivariate copulas for a more coherent representation of flood behavior, enhancing the robustness of hydrological analyses.

Grimaldi & Serinaldi (2006) introduced duration in hydrological estimation using three-variable copulas, applying both symmetric and asymmetric copulas from the Archimedean family to model events in the Kanawha River. They demonstrated that asymmetric copulas are more flexible in representing heterogeneous dependencies between flow, volume, and duration. Wong *et al.* (2010) used trivariate copulas for drought characterization, while Zhang & Singh (2007) applied the Gumbel-Hougaard copula for trivariate flood frequency analysis, showing improved performance over the trivariate normal distribution and highlighting that return periods for joint extremes are often lower when copulas are used. Ganguli & Reddy (2013) applied a trivariate copula-based approach to flood risk analysis in the Delaware River basin, concluding that multivariate return periods are critical for capturing the simultaneous occurrence of flood characteristics. Similarly, Bezak *et al.* (2014) and Orcel *et al.* (2021) extended trivariate models to include sediment concentration and coastal variables, respectively, reinforcing their applicability across hydrological contexts.

Despite these advantages, trivariate frequency analysis presents several limitations that must be considered to ensure its effective application. First, it requires long and high-quality hydrological records to produce statistically robust estimates, particularly for low-probability events. The complexity of the dependence structure among three variables increases computational burden and complicates model calibration, especially when evaluating multiple copula families. These models are also sensitive to outliers, which can distort marginal fitting and joint estimation. Additionally, defining input variables – especially event duration – from measured data is not straightforward. In many studies, duration is visually identified based on the analyst's discretion, introducing subjectivity and limiting reproducibility. Furthermore, when variables are selected independently rather than jointly, the observed correlation between them can decrease significantly, undermining the copula's ability to capture their true dependence. Interpreting and communicating trivariate results – such as joint return periods – also remains challenging for stakeholders unfamiliar with probabilistic methods. To address these issues, joint variable selection, rigorous data preprocessing, outlier detection, and the use of flexible copula structures are recommended. Acknowledging and managing these limitations is essential to apply trivariate models responsibly and effectively.

To overcome these limitations, this article develops a methodology based on both asymmetric and symmetric trivariate copulas, incorporating duration as a primary variable during data selection and event estimation to preserve the natural correlation structure between variables. The Gumbel-Hougaard and Clayton copulas from the Archimedean family are evaluated, with marginal distributions estimated using two-population probability functions via maximum likelihood and genetic algorithms (GAs). A hybrid peak selection approach combining the annual maxima and threshold methods is implemented to reduce subjectivity in event definition. Unlike traditional approaches that assume fixed hydrograph shapes, the proposed system allows any user-defined shape to be applied to the estimated values of peak flow, volume, and duration. The methodology is validated through a case study at the La Angostura dam in Mexico, demonstrating its practical applicability. By maintaining the joint behavior of variables and minimizing arbitrary assumptions, the proposed approach improves the statistical consistency and reliability of design hydrographs for flood risk assessment and reservoir management.

MATERIALS AND METHODS

Study area description

The area chosen for this study is the Yaqui River basin, located in the state of Sonora, in northern Mexico. This river, one of the most important in the region, plays a crucial role in supplying water for agricultural, urban, and ecological activities. Specifically, the study focuses on the estimation and review of design hydrographs for the General Lázaro Cárdenas Dam,

known as ‘La Angostura’, a key infrastructure for water management and regulation of the river. The basin has a predominantly arid and semi-arid climate, with rainfall concentrated during the summer (monsoon season) and a runoff regime that can be highly variable. Additionally, in the higher areas of the basin, significant runoff occurs due to snowmelt following the winter season, which increases the seasonal variability of flows. Average annual precipitation ranges from 300 to 800 mm, depending on altitude and location within the basin, presenting significant challenges for the design and operation of hydraulic structures. Figure 1 presents the location and basin of the La Angostura Dam.

Dataset

The contributing area of the General Lázaro Cárdenas Dam basin, known as ‘La Angostura’, is 18,100 km². For this study, daily mean inflow records to the reservoir were provided by the Comisión Federal de Electricidad (CFE), which operates the dam. The dataset spans from 1942 to 2006, excluding the years 1987–1993, during which the hydrometric station was out of service and no measurements were taken. This results in 58 years of valid data. The use of daily mean discharges ensures temporal consistency with the runoff event scale, allowing appropriate association with event duration over 24-h periods. The data were subjected to quality control procedures, including the identification of missing or inconsistent values and statistical testing to confirm homogeneity and independence. This dataset provides a robust foundation for hydrological characterization and supports the reliability of the frequency analysis and design hydrograph estimation.

Method of data analysis

Joint data selection

The first step in trivariate copula modeling is the selection of data in a way that preserves the natural correlation among variables. In hydrograph estimation, summarized hydrometric data – such as daily discharge records – are commonly used. A

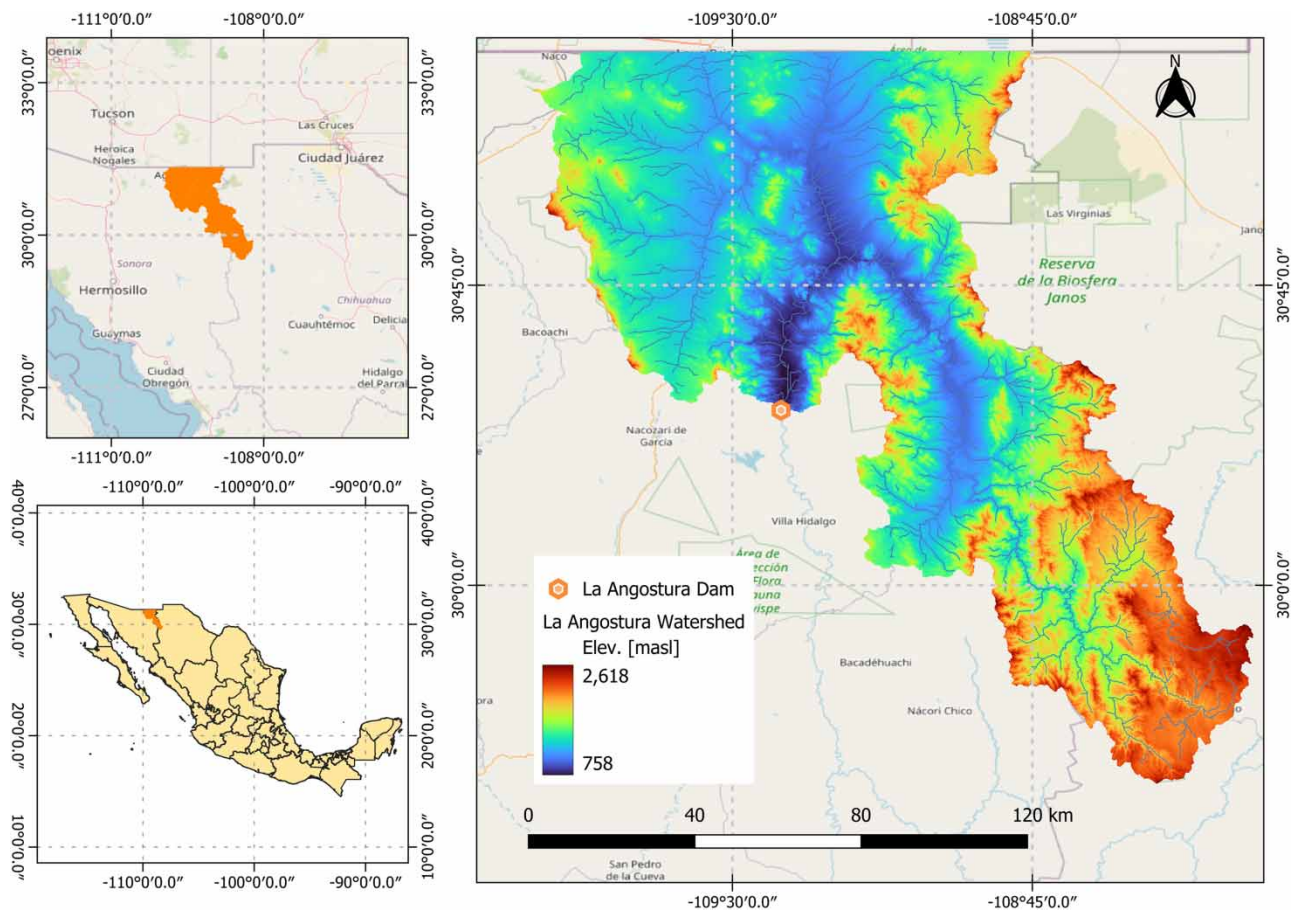


Figure 1 | Location of the Angostura dam within the Yaqui River Basin in Sonora, Mexico.

typical approach involves identifying the annual maxima for flow and volume and associating them with a representative duration. However, this method does not guarantee that the interdependence between variables is maintained.

To overcome this limitation, a hybrid joint data selection method was applied, focusing on high-magnitude flood events and their corresponding durations. This method combines techniques from both the annual maximum series (AMS), which selects the maximum value of a variable for each year, and the peaks over threshold (POT) method (Stamatatou *et al.* 2018), which identifies values exceeding a predefined threshold to capture extreme events. The combination of AMS and POT improves the accuracy and reliability of the selected data for trivariate copula-based hydrological modeling.

Hydrographs with variable durations were identified as follows:

- (1) First, the annual maximum streamflow was selected for each year of the dataset, forming the population of peak discharges (AMS).
- (2) Duration was then prioritized as a key variable to define. Using the POT method, a threshold was established such that the smallest of the previously identified annual peak flows exceeded this threshold.
- (3) Events exceeding the threshold were delimited by the time intervals during which discharge remained above the threshold, thus defining the start and end of each event, which corresponds to its duration.
- (4) The volume was computed as the integral of the daily discharge over the defined duration.

This procedure ensured that each year with a recorded maximum flow was associated with a unique flood event, characterized by a triplet: maximum discharge, volume, and duration. These triads formed the basis for trivariate frequency analysis. In some cases, particularly for the event corresponding to the year with the smallest annual maximum discharge, the resulting duration may be as short as one day. This outcome is a consequence of the threshold-based method and may be considered a limitation. However, for the majority of selected events, the duration obtained is consistent with the temporal structure of typical flood events, ensuring coherence across the dataset.

Figure 2 illustrates 3 years of streamflow data. The red dots represent the annual maximum discharges, while the horizontal line denotes the threshold. The duration of each event is bounded by the first and last exceedances of the threshold.

Marginal probabilities

Before analyzing the joint behavior of two or more hydrological variables, it is essential to estimate their marginal probability distributions. These functions provide the probability that a given variable does not exceed a specified threshold and enable the transformation of observed data into a standardized uniform scale. This standardization is a prerequisite for applying copula-based models, which require input variables to be expressed in terms of their cumulative distribution functions

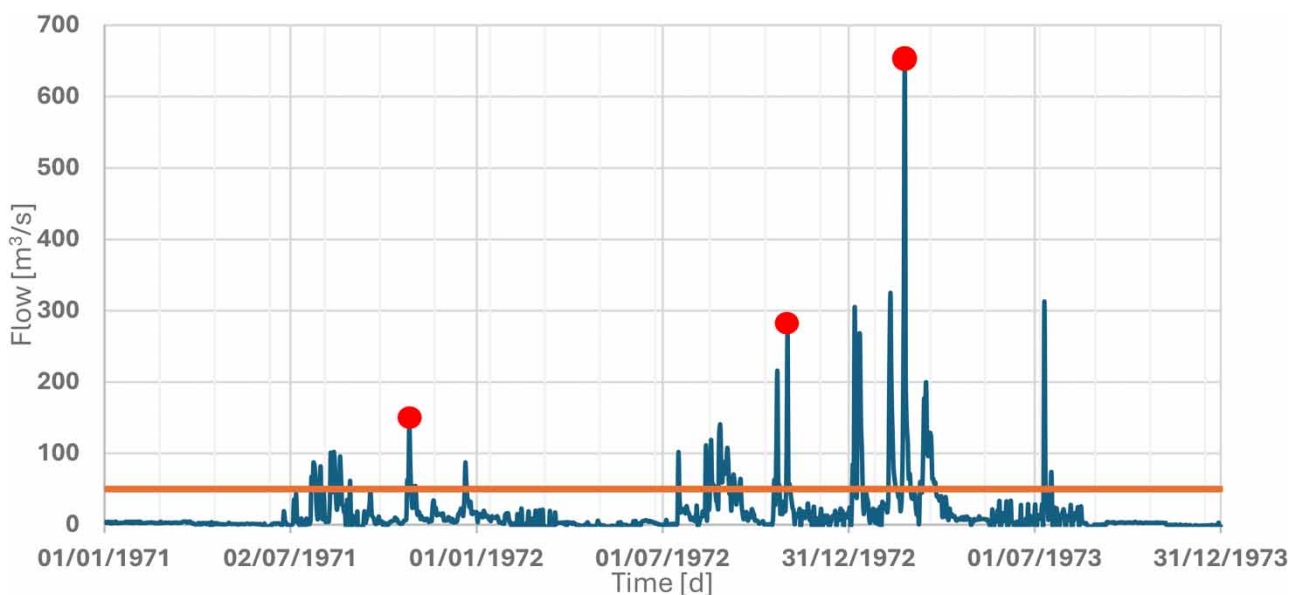


Figure 2 | Observed average daily flows for 3 years, in red is highlighted the maximum annual flow, in orange the threshold proposed.

(CDFs) rather than raw values. Consequently, marginal estimation constitutes the foundation for any multivariate frequency analysis.

This study applies one- and two-population probability functions using maximum likelihood estimation with GAs, considering that runoff events in the La Angostura dam area are of two types. The employed functions include Gumbel, Gumbel-Gumbel, Weibull, Weibull-Weibull, Gamma, Gamma-Gamma, Gumbel-Weibull, Weibull-Gamma, and Gumbel-Gamma, as detailed in Table 1. These adjustments were made using software developed by the Engineering Institute at the National Autonomous University of Mexico (UNAM II). The computation of marginal distributions is standard practice in hydrology. However, the use of two-population functions with GA – while appropriate for capturing mixed event types – may pose challenges for design engineers due to their increased complexity.

Trivariate copulas

A probability copula is a multivariate function $C:[0, 1]^d \rightarrow [0, 1]$ that links the marginal CDFs of d random variables $U_1, U_2, \dots, U_d$, where each U_i has a uniform distribution on the interval $[0, 1]$ (Sklar 1959; Joe 1997; Nelsen 1999).

For a vector of random variables $U = (U_1, U_2, \dots, U_d)$ with CDFs $F_1, F_2, \dots$, the copula C is defined as

$$C(F_1(u_1), F_2(u_2), \dots, F_d(u_d)) = P(U_1 \leq F_1(u_1), U_2 \leq F_2(u_2), \dots, U_d \leq F_d(u_d)) \quad (1)$$

where P denotes probability and U_i are the uniformly distributed random variables. The copula captures solely the dependence structure between the random variables, independently of their marginal distributions.

A wide variety of copula families have been proposed, as established by Sklar's framework, and are commonly classified into categories such as elliptical and Archimedean. Schweizer & Sklar (1983) demonstrated that Archimedean copulas for two random variables u and v can be constructed using a generator function φ , which defines the dependence structure. Genest & Favre (2007) formalized procedures for applying copulas to hydrological modeling.

Trivariate copulas are extensions that allow for modeling the joint dependence of three variables, making them particularly suited for hydrological applications involving peak discharge, volume, and duration. Unlike traditional multivariate distributions, copulas allow flexibility by separating marginal behavior from joint dependence. The strength of dependence is

Table 1 | Marginal CDF used to estimate the marginal probability for q, v, d

Name	CDF
Gumbel [Gum]	$F_X(x) = \exp\{-\exp(-\alpha(x - \beta))\}$
Weibull [Wei]	$F_X(x) = 1 - \exp\left[-\left(\frac{x}{\beta}\right)^\alpha\right]$
Gamma [Gam]	$F_X(x) = \int_0^x \frac{1}{\alpha \Gamma(\beta)} \left(\frac{x}{\beta}\right)^{\alpha-1} \exp\left[-\left(\frac{x}{\beta}\right)^\alpha\right] dx$
Gumbel-Gumbel [Gum-Gum]	$F_X(x) = z \exp\{-\exp(-\alpha_1(x - \beta_1))\} + (1 - z) \exp\{-\exp(-\alpha_2(x - \beta_2))\}$
Gumbel-Weibull [Gum-Wei]	$F_X(x) = z \exp\{-\exp(-\alpha_1(x - \beta_1))\} + (1 - z) \left\{1 - \exp\left[-\left(\frac{x}{\beta_2}\right)^{\alpha_2}\right]\right\}$
Gumbel-Gamma [Gum-Gam]	$F_X(x) = z \exp\{-\exp(-\alpha_1(x - \beta_1))\} + (1 - z) \left\{\int_0^x \frac{1}{\alpha_2 \Gamma(\beta_2)} \left(\frac{x}{\beta_2}\right)^{\alpha_2-1} \exp\left[-\left(\frac{x}{\beta_2}\right)^{\alpha_2}\right] dx\right\}$
Weibull-Weibull [Wei-Wei]	$F_X(x) = z \left\{1 - \exp\left[-\left(\frac{x}{\beta_1}\right)^{\alpha_1}\right]\right\} + (1 - z) \left\{1 - \exp\left[-\left(\frac{x}{\beta_2}\right)^{\alpha_2}\right]\right\}$
Weibull-Gamma [Wei-Gam]	$F_X(x) = z \left\{1 - \exp\left[-\left(\frac{x}{\beta_1}\right)^{\alpha_1}\right]\right\} + (1 - z) \left\{\int_0^x \frac{1}{\alpha_2 \Gamma(\beta_2)} \left(\frac{x}{\beta_2}\right)^{\alpha_2-1} \exp\left[-\left(\frac{x}{\beta_2}\right)^{\alpha_2}\right] dx\right\}$
Gamma-Gamma [Gam-Gam]	$F_X(x) = z \left\{\int_0^x \frac{1}{\alpha_1 \Gamma(\beta_1)} \left(\frac{x}{\beta_1}\right)^{\alpha_1-1} \exp\left[-\left(\frac{x}{\beta_1}\right)^{\alpha_1}\right] dx\right\} + (1 - z) \left\{\int_0^x \frac{1}{\alpha_2 \Gamma(\beta_2)} \left(\frac{x}{\beta_2}\right)^{\alpha_2-1} \exp\left[-\left(\frac{x}{\beta_2}\right)^{\alpha_2}\right] dx\right\}$

governed by an association parameter θ , whose interpretation and admissible range depend on the copula family used (Nelsen 1999; Salvadori *et al.* 2007).

Trivariate symmetric copulas

Symmetric trivariate copulas describe the joint behavior of three random variables under the assumption that their dependence structure remains unchanged under any permutation of the variables (Joe 1997). This means that the copula treats the relationship among all pairs equally, regardless of the order in which the variables are arranged (Grimaldi & Serinaldi 2006).

$$C(u_1, u_2, \dots, u_n) = \varphi^{-1}(\varphi(u_1) + \dots + \varphi(u_n)) \quad (2)$$

where φ is the generating function of the copula.

These copulas are particularly useful in systems where the interaction among variables is expected to be balanced and symmetric. The associated dependence parameter assumes equal strength of association among all variable pairs, implying identical correlation between each pair of the three variables (Wong *et al.* 2008). However, in many hydrological applications, this assumption is often unrealistic. Symmetric copulas may underestimate the strength of dependence between the most strongly correlated variables and overestimate it for weaker relationships (Wong *et al.* 2010).

The Archimedean copula family (Nelsen 1999) is widely used in hydrology due to its simplicity, analytical tractability, and ability to model multivariate extreme events with flexible dependence structures.

Trivariate asymmetric copulas

Trivariate asymmetric copulas offer a significant advantage over their symmetric counterparts by allowing for more flexible modeling of variable dependence. Unlike symmetric copulas, which assume equal correlation among all variable pairs, asymmetric structures can assign stronger dependence between a specific pair of variables while treating the third variable differently. This results in a more realistic representation of the underlying dependence structure, especially in hydrological applications where intervariable correlations are often heterogeneous (Grimaldi & Serinaldi 2006; Bezak *et al.* 2014).

Trivariate asymmetric copulas consist of two interconnected bivariate copulas. The first of these copulas links the two variables with the strongest correlation u_1 and u_2 , while the second copula connects this bivariate pair with the third variable u_3 . It is important to note that this approach is particularly useful in situations where the correlation between u_1 and u_2 significantly exceeds the correlation between u_1 and u_3 , as well as between u_2 and u_3 (Serinaldi & Grimaldi 2007; Wong *et al.* 2010; Ganguli & Reddy 2013).

The trivariate asymmetric copula is invaluable for modeling complex variable interactions that do not fit a symmetric model. It captures specific relationships between variables with greater precision, enhancing decision-making and providing a deeper understanding of the analyzed processes (Grimaldi & Serinaldi 2006).

A trivariate asymmetric copula is given by the form:

$$\begin{aligned} C(u_1, u_2, \dots, u_n) &= C_1(u_n, C_2(u_{n-1}, \dots, C_{n-1}(u_2, u_1) \dots)) \\ C(u_1, u_2, u_3) &= C_1(u_3, C_2(u_2, u_1)) = \varphi_1^{-1}(\varphi_1(u_3) + \varphi_1 \circ \varphi_2^{-1}(\varphi_2(u_2) + \varphi_2(u_1))) \end{aligned} \quad (3)$$

In various hydrological studies, the Archimedean family of copulas has been used (Salvadori & Michele 2004; Zhang & Singh 2007; Bačová & Halmova 2014). Arquimedean family copulas are presented in Table 2.

Trivariate asymmetric copulas, as opposed to their symmetric counterparts, feature the presence of two parameters, with each parameter corresponding to one of the bivariate copulas that form it. θ_2 represents the variables exhibiting stronger dependence, while θ_1 corresponds to the bivariate copula involving θ_2 and the variable exhibiting less dependence with the other two.

For the copula to exist, it must satisfy the condition that $\theta_2 > \theta_1$. If $\theta_2 = \theta_1$, the copula transitions from being asymmetric to the symmetric case.

Association parameter θ

To model a trivariate copula, the association parameter must be estimated. Various methods exist for this purpose, with the simplest involving the use of dependence measures such as Kendall's tau and Spearman's rho (Genest & MacKay 1986;

Table 2 | Symmetric and asymmetric trivariate copulas for the Archimedean family

Name	Symmetric $C(u_1, u_2, u_3) =$	Asymmetric $C_{\theta_1, \theta_2, \theta_3}(u_1, u_2, u_3) =$
Gumbel-Hougaard	$\exp\{-(\ln u_1)^\theta + (\ln u_2)^\theta + (\ln u_3)^\theta\}$	$\exp\{-[(\ln u_1)^{\theta_2} + (\ln u_2)^{\theta_2}]^{\theta_1/\theta_2} + (\ln u_3)^{\theta_1}\}^{1/\theta_1}$
Clayton	$\{u_1^{-\theta} + u_2^{-\theta} + u_3^{-\theta} - 2\}^{-1/\theta}$	$\{(u_1^{-\theta_2} + u_2^{-\theta_2} - 1)^{\theta_1/\theta_2} + u_3^{-\theta_1} - 1\}^{1/\theta_1}$
Frank	$-\frac{1}{\theta} \ln\left(1 + \frac{(\exp(-\theta u_1) - 1)(\exp(-\theta u_2) - 1)(\exp(-\theta u_3) - 1)}{(\exp(-\theta) - 1)^2}\right)$	$-\theta_1^{-1} \log\{1 - (1 - e^{-\theta_1})^{-1}(1 - [1 - (1 - e^{-\theta_2})^{-1}x(1 - e^{-\theta_2 u_1})]^{1/\theta_2})^{\theta_1/\theta_2}(1 - e^{-\theta_1 u_3})\}$

(Chowdhary 2009), and their relationship with the parameter. Another approach, presented by Genest *et al.* (1995), utilizes pseudo-maximum likelihood estimation. For this, the probability density function (PDF) for trivariate copulas is required, which is given by

$$c(u_1, u_2, u_3) = \frac{\partial^3 C(u_1, u_2, u_3)}{\partial u_1 \partial u_2 \partial u_3} \quad (4)$$

The likelihood function $V(\theta)$ of the copula function C is expressed as

$$V(\theta) = \prod_{k=1}^{k=n} c(F_u(u_k), \dots, F_v(v_k)) \quad (5)$$

Given c as the density function of the copula C ; u_k, v_k are the k -th random variables u and v . The marginal functions are often multiplied by $n/(n+1)$ to prevent them from tending to zero (Genest *et al.* 1995).

Trivariate goodness-of-fit

To evaluate the performance of trivariate joint probability estimates obtained using the Clayton and Gumbel-Hougaard copulas, the observed cumulative frequency (OCF) is compared against the fitted copula values using goodness-of-fit tests. The OCF is computed with the Gringorten formula (Gringorten 1963), which estimates empirical probabilities for ordered data as

$$P_k = \frac{k - 0.44}{n + 0.12} \quad (6)$$

This empirical distribution provides a benchmark for assessing how well the copula-based model replicates the observed joint probability structure.

To determine the adequacy of the fitted copulas, statistical tests, such as the Kolmogorov-Smirnov (KS) test (Zhang & Singh 2007) and the Anderson-Darling (AD) test (Ane & Kharoubi 2003), are applied in their multivariate forms. These tests are essential for selecting the copula that most accurately captures the dependence structure among the variables.

The test statistics are calculated as follows:

$$KS = \max |C(u, v, d) - OCF| \quad (7)$$

$$AD = \max \left(\frac{|OCF - C(u, v, d)|}{\sqrt{C(u, v, d)[1 - C(u, v, d)]}} \right) \quad (8)$$

Selecting an inappropriate copula can lead to a misrepresentation of the dependence structure among the variables, particularly in the tails of the distribution.

Trivariate return period

Once the copula parameters have been estimated, the next step in modeling multivariate extreme events is to define an appropriate return period. Unlike the univariate case, the concept of return period in a multivariate framework must incorporate the joint behavior of variables. The trivariate return period (Tr) is defined using the joint probability that three variables simultaneously exceed specified thresholds. This definition is essential when dealing with interdependent variables, such as peak discharge, volume, and duration in flood modeling. If the variables are independent, their joint exceedance probability is simply the product of their marginal exceedance probabilities. However, when dependence exists – as is typical in hydrological events – copula models must be used to account for this structure.

Salvadori *et al.* (2011) and Gräler *et al.* (2013) presented and studied two cases of joint return periods, which Bezak *et al.* (2014) defines as primary and secondary joint return periods for trivariate cases. The first case, the OR joint return period (Tr_{OR}) indicates that one of the three variables is exceeded ($U > u$ or $V > v$ or $D > d$). Salvadori & Michele (2004) utilized the Kendall distribution function to define the ‘secondary return period’ (Tr_{KEN}). The primary return period predicts that a critical event is expected to occur once within a given time interval, providing an average forecast. In contrast, the secondary return period provides the average time between the occurrences of two supercritical events (Ganguli & Reddy, 2013).

$$Tr_{OR} = \frac{1}{1 - C} \quad (9)$$

The second case is the Kendall’s return period given by

$$Tr_{KEN} = \frac{1}{1 - P_{KEN}} = \frac{1}{1 - Kc(t)} \quad (10)$$

where $Kc(t)$ is the Kendall’s distribution of probability associated to the copula.

The return period is meaningful because the critical layer associated with the Kendall’s return period partitions Rd into three regions: subcritical region ($Rt <$), supercritical region ($Rt >$), and critical layer (Salvadori *et al.* 2011). An advantage of this approach, compared with the OR methodology, is that all realizations lying over the critical layer, which is defined with a selected t value, have the same Kendall’s return period value (Salvadori *et al.* 2011; Volpi & Fiori 2014).

The Kendall distribution function of the real variable t within the closed interval $[0,1]$ (Genest *et al.* 1995) is expressed as

$$K(t) = t - \frac{\varphi(t)}{\varphi'(t)} \quad (11)$$

where $\varphi(t)$ corresponds to the generating function of the copula of interest.

Defining the hydrograph shape

Once the peak flow, volume, and duration corresponding to the desired joint return period have been determined, various methods can be employed to reconstruct the hydrograph of the maximum annual flood. These include approaches based on probability density functions for synthetic shapes or the use of representative historical flood hydrographs. It is essential that the selected shape preserves the relationships among the variables estimated by the copula, as represented by the joint probability isolines. Therefore, the hydrograph must be constructed in a way that respects the dependence structure, requiring a formulation that operates in at least a bivariate context.

To address this, a flexible system of equations has been proposed (Equations (12) and (13)), which enables the hydrograph shape to be adapted to the values associated with the selected return period. This system allows for nearly any hydrograph form, provided that the input variables remain correlated:

$$a \sum f_i + b \sum f_i Q_i \Delta t = V \quad (12)$$

$$a + b Q_p \Delta t = Q \Delta t \quad (13)$$

In this formulation, f_i represents the relative shape coefficients associated with each discretized flow value Q_i , expressed as a function of the peak flow Q_p . The variable Δt denotes the uniform time step used to define the hydrograph. Q and V

correspond to the design discharge and volume values associated with the joint return period T_r . The coefficients a and b are unknowns solved from the system to ensure that the resulting hydrograph simultaneously satisfies both the total volume and peak discharge constraints. Once the system is resolved, the new flow series Q_i can be computed, defining the complete hydrograph shape.

This methodology integrates marginal estimation, trivariate copula modeling, joint return period analysis, and hydrograph shaping into a unified framework that prioritizes event duration in the design of flood hydrographs. By explicitly incorporating duration as a core variable – alongside peak flow and volume – the approach preserves the natural dependence structure among these variables and avoids arbitrary assumptions. The resulting hydrographs are statistically consistent and physically representative, making them particularly useful for flood routing through reservoirs, flood risk assessment, spillway design, emergency planning, and other applications in hydraulic infrastructure management. Furthermore, the method's flexibility in hydrograph reconstruction ensures adaptability to a range of design scenarios and planning needs.

RESULTS

Event selection and variable characterization

To emphasize the role of duration in trivariate modeling, the proposed methodology was applied to generate inflow hydrographs for the La Angostura dam in Mexico. The analysis was based on 58 years of average daily flow (ADF) records spanning from 1942 to 2006, excluding the period from 1987 to 1993 due to missing data. From these records, the key hydrological variables – peak flow (Q , m^3/s), volume (V , $\text{m}^3/\text{s}\cdot\text{d}$), and duration (D , days) – were identified using the procedure outlined in the second section.

For each year, the annual maximum peak flow was extracted, and a threshold value of $50 \text{ m}^3/\text{s}$ was established. This threshold was defined to ensure that even the lowest of the annual peak flows was associated with a clearly delineated flood event. Based on this criterion, the start and end dates of each event were determined, allowing the calculation of event duration and corresponding total volume. The selected events exhibit a wide range of durations and magnitudes. A statistical summary of these variables is presented in Table 3, and their temporal profiles are illustrated in Figures 3 and 4.

Dependence analysis among variables

Using the selected values of peak flow (Q), volume (V), and duration (D), the dependence structure between variable pairs was analyzed to characterize their interrelationships and mutual influence. Understanding these dependencies is essential for improving the accuracy of joint modeling and the reliability of flood estimation. The strength and nature of dependence were assessed using three statistical measures: Pearson's linear correlation coefficient, Kendall's tau, and Spearman's rho. These indicators capture both linear and rank-based associations, providing a robust evaluation of intervariable relationships. The results are summarized in Table 4.

The (Q , V) pair exhibits the strongest dependence among the three combinations, indicating a close relationship between peak flow and volume. In contrast, the (Q , D) pair shows relatively weak correlation, which may introduce challenges when modeling their joint behavior using copulas. The (V , D) relationship, however, appears naturally coherent and directly proportional, as both variables are closely linked through the definition of flood volume. It is also important to note that the interaction between Q and D presents a higher degree of randomness, suggesting a more stochastic relationship that may be difficult to capture with simple dependence structures.

For cases like this, where one pair exhibits stronger dependence compared with the other two, asymmetric copulas represent an alternative to modeling with symmetric copulas (Bezak *et al.* 2014). The pair with the highest dependency corresponds to the association parameter θ_2 . In this work, we designate $u_1 = q$ and $u_2 = v$; hence, $u_3 = d$ represents d ,

Table 3 | Statistics for the flow, volume, and duration variables selected

Statistic	Q	V	D
Mean	459.964	1,215.087	7.862
Std. Dev	349.673	1,027.336	2.775
Kurtosis	2.760	6.722	0.104
Asymmetry	1.541	2.268	−0.036

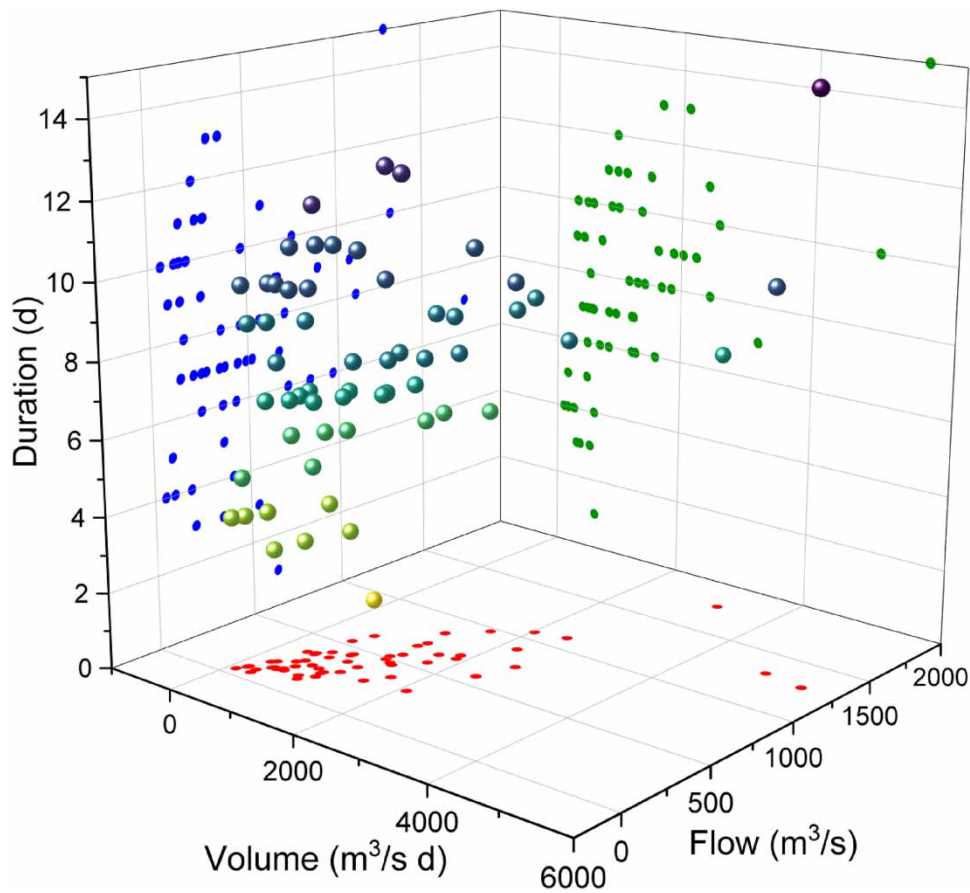


Figure 3 | Thirds Q , V , D selected, it is evident that the major observed events correspond to durations between 5 and 10 days.

and the relationship between $C(q, v)$ and d corresponds to θ_1 . The duration appears to play a secondary role in the estimation since the variables q and v have a higher correlation with each other. However, instead of being a drawback, this presents an opportunity to handle duration independently, thereby prioritizing it in the subsequent steps of the methodology.

Marginal probability estimation

To determine the association parameter using the method proposed by [Genest et al. \(1995\)](#), it is essential to accurately estimate the marginal probabilities of the variables. Proper marginal fitting ensures that the copula-based model reliably captures the dependence structure inherent in natural hydrological processes. Although copulas are mathematically capable of modeling independent variables, incorporating actual interdependence is critical for realistic modeling.

For the variables of peak flow and volume, marginal distributions were selected based on multiple univariate goodness-of-fit criteria, including the root mean square error (RMSE), the KS test, the AD test, the Akaike information criterion (AIC) ([Akaike 1974](#)), the Bayesian information criterion (BIC) ([Schwarz 1978](#)), their RMSE-modified versions (AIC mod, BIC mod), and the Pham information criterion (PIC) ([Pham 2019](#)). Special emphasis was placed on tests focused on probability accuracy, rather than value matching, given that copulas operate on probability space. As a result, the selected functions may not yield the best scores across all criteria, but were prioritized for their compatibility with copula modeling. The results of the fitting process are summarized in [Table 5](#).

For the duration variable, special consideration is required due to its fundamentally different nature in extrapolation. Although goodness-of-fit tests indicated that the Gumbel-Gumbel or Weibull-Weibull distributions best represent the empirical data, these functions may produce unrealistic extrapolations in the tails. Unlike flow and volume, duration does not exhibit a monotonic increase with probability, and its behavior can distort the trivariate dependency structure if not carefully managed.

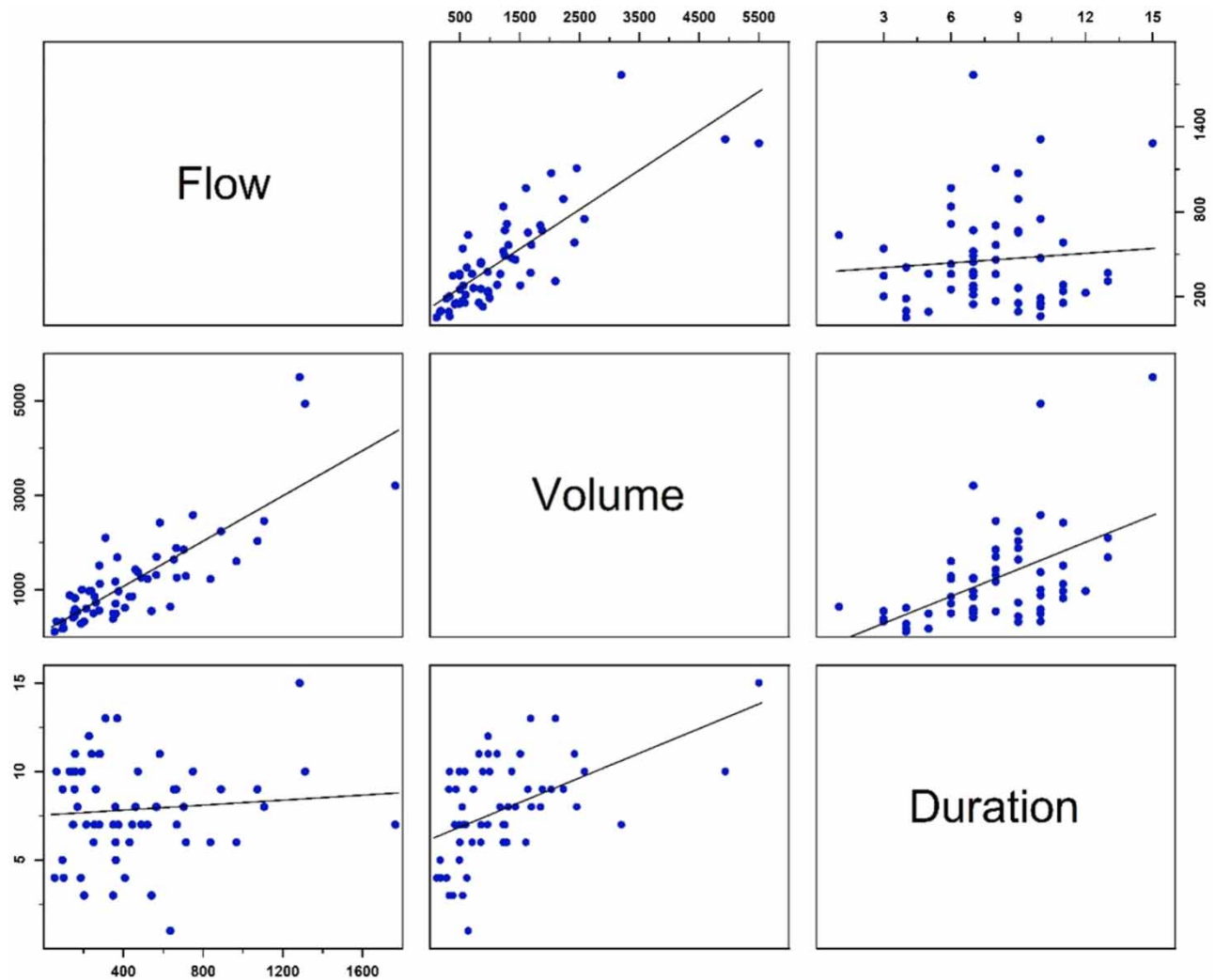


Figure 4 | Scatterplot between the three pairs of data (q, v), (v, d), and (q, d).

Table 4 | Dependence between three pairs of data

Variable pairs	Pearson's rho	Kendall's tau	Spearman's rho
Q, V	0.8153	0.6237	0.8066
V, D	0.5149	0.3521	0.5044
Q, D	0.0893	0.0121	0.0019

To address this, a tailored approach was applied to the duration variable. Rather than selecting the best-fitting distribution, the goal was to stabilize the marginal tail and prevent excessive growth in extrapolated values. This was achieved by intentionally selecting a single-population distribution – in this case, the Weibull distribution – with more stable tail behavior. Under this approach, two-population models were discarded for duration due to their poor performance in extrapolation control. The parameters of the final selected marginal distributions for all variables are presented in [Table 5](#).

When the range of variables is small, such as in the case of duration, it is better to maintain single-population functions. In contrast, if the range is wide, it is advisable to use two-population functions, always in accordance with the quality of the marginal probability fit due to the nature of the variable.

Table 5 | Parameters and goodness-of-fit tests values obtained for the specified functions Gumbel-Gumbel for flow and volume and Weibull for duration

	Parameter	Gu-Gu	We-We	Ga-Ga	Gu-We	Gu-Ga	We-Ga	Gu	We	Ga
Flow	α_1	0.0111	327.285	224.0873	0.0038	0.0034	894.6028	0.0037	500.174	265.829
	β_1	168.252	1.8209	2.8437	499.300	567.8405	2.0887	302.592	1.3291	1.7303
	α_2	0.0038	761.22	77.39	2.1302	3.0849	122.1386	0	0	0
	β_2	502.387	1.769	3.1851	257.933	88.9181	2.6817	0	0	0
	p	0.444	0.574	0.56	0.554	0.402	0.271	0	0	0
	RMSE	57.631	47.607	42.926	61.172	44.487	50.699	98.873	68.637	64.864
	AIC mod	474.49	458.11	440.32	481.41	444.46	465.41	770.57	1,452.25	487.99
	BIC mod	480.27	456.23	446.1	487.19	450.24	463.53	774.48	732.14	492.11
	PIC	176,033	120,126	97,667	198,331	104,899	136,236	547,447	263,822	235,611
	KS	0.0398	0.0397	0.0424	0.0407	0.0469	0.0458	0.0879	0.0656	0.0573
	AD	0.132	0.128	0.102	0.144	0.1	0.119	0.635	0.348	0.21
	AIC	824.75	828.12	823.79	824.74	823.96	828.12	825.31	820.7	819
	BIC	833.9	837.26	832.94	833.89	833.11	837.26	829.22	824.6	822.91
Volume	α_1	0.0034	2,567.72	1,102.07	0.0007	0.0008	1,109.111	0.0012	1,287.92	868.597
	β_1	504.058	1.5035	2.3145	2,511.80	1,667.392	1.6934	752.730	1.1865	1.3989
	α_2	0.0013	1090.06	415.156	2.3788	1.7062	1,042.464	0	0	0
	β_2	1,313.301	1.7216	2.3687	432.281	1,084.821	2.8404	0	0	0
	p	0.502	0.167	0.153	0.086	0.136	0.893	0	0	0
	RMSE	323.217	329.23	211.437	212.814	308.022	265.008	416.785	325.169	326.986
	AIC mod	674.5	651.4	625.27	626.03	668.92	657.25	937.47	1,813.13	675.63
	BIC mod	680.29	649.52	631.06	631.81	674.7	655.37	941.37	912.58	679.75
	PIC	5,536,869	3,364,923	2,369,402	2,400,354	5,028,501	3,722,158	9,727,739	5,921,167	5,987,531
	KS	0.0485	0.0648	0.0539	0.0595	0.0562	0.0644	0.0934	0.0545	0.0595
	AD	0.224	0.204	0.15	0.163	0.24	0.205	0.68	0.549	0.319
	AIC	940.13	944.77	937.38	937.4	938.63	944.77	940.67	936.43	935.75
	BIC	949.28	953.91	946.53	946.55	947.78	953.91	944.58	940.33	939.65
Duration	α_1	0.4617	9.2246	1.5574	0.5866	0.5961	8.7996	0.4617	8.7914	0.9816
	β_1	4.2874	5.0641	3.4886	7.5266	7.8001	3.352	6.6118	3.0986	8.0094
	α_2	0.573	8.1068	0.5768	4.8225	3.6975	0.312	0	0	0
	β_2	7.8102	2.3357	14.9799	3.4894	1.5728	3.5088	0	0	0
	p	0.288	0.494	0.245	0.716	0.69	0.981	0	0	0
	RMSE	0.402	0.43	0.408	0.842	0.403	0.447	0.891	0.451	0.956
	AIC mod	-101.63	-87.83	-99.79	2.26	-101.25	-83.41	224.34	286.55	-1.27
	BIC mod	-95.85	-89.71	-94.01	6.55	-95.46	-85.29	228.24	149.29	2.85
	PIC	14	15	14	13	14	16	47	13	53
	KS	0.0961	0.0923	0.7444	0.1231	0.1014	0.0953	0.1494	0.1019	0.1518
	AD	0.514	0.465	3.193	0.237	0.539	0.493	1.272	0.542	0.623
	AIC	292.63	293.79	292.57	71.31	292.72	293.79	295.63	286.84	296.62
	BIC	301.77	302.93	301.71	57.83	301.87	302.93	299.53	290.74	300.53

Figure 5 shows the selected marginal probability distributions for Angostura Dam with plots. The duration for high return periods is established to range from 10 to 15 days.

Copula model

Once the marginal probabilities and dependencies among the three pairs of variables are determined, the next step involves estimating copula parameters for symmetric and asymmetric cases using the Genest method. This approach is essential for accurately modeling the probabilistic relationships between variables. The Genest method offers a reliable framework for parameter estimation in both cases.

The KS test assesses the goodness-of-fit between the empirical data distribution and the theoretical distribution assumed by the copula model, providing a measure of how well the copula fits the observed data. The AD test offers a more sensitive

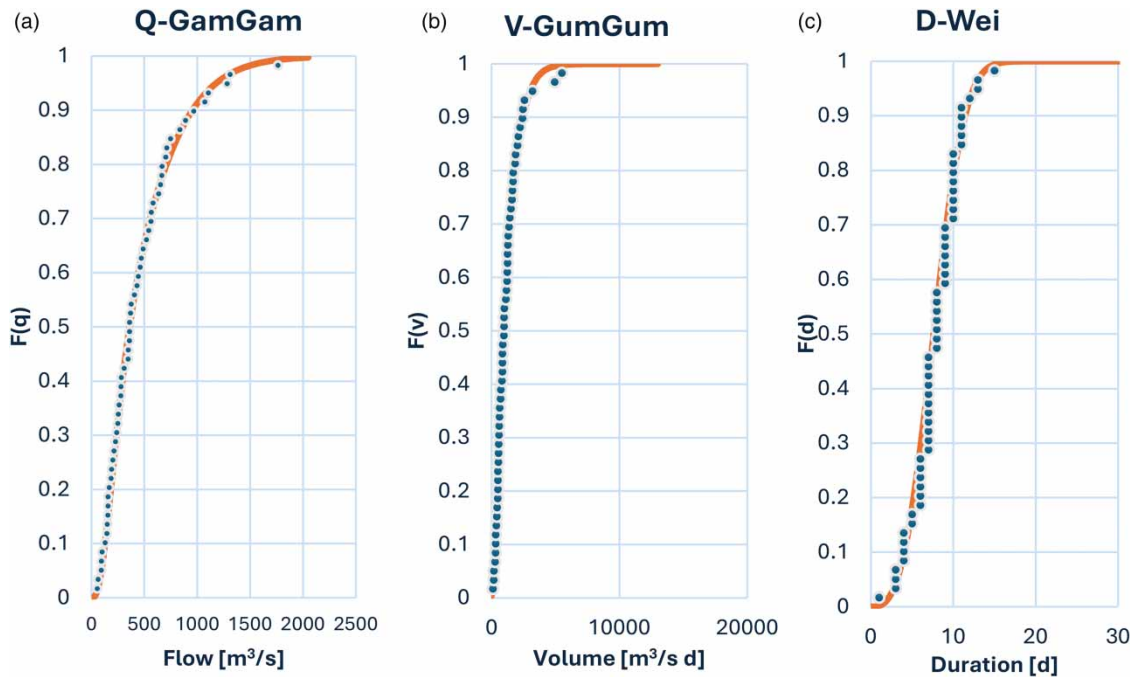


Figure 5 | Gamma-Gamma, Gumbel-Gumbel, and Weibull probability plots of flow (a), volume (b), and duration (c).

measure, particularly for detecting deviations in the tail behavior of the distribution, and is used to evaluate the goodness-of-fit of a statistical model to the data. By comparing the p -values from both tests for all candidate copulas, the copula with the highest p -value (indicating the best fit) is chosen as the most suitable model.

Both the Clayton and Gumbel-Hougaard copulas show that the asymmetric copula provides a better fit than the symmetric copula. This indicates that incorporating asymmetry in the dependency structure results in a more accurate model of the observed data. Among the two asymmetric copulas, the Clayton copula performs best, offering the most precise representation of the trivariate joint probability distribution. This highlights the Clayton copula's effectiveness in capturing the data's dependency structure, making it the preferred choice for modeling in this context.

The association parameters obtained for the asymmetric copulas require that θ_2 must be greater than θ_1 , as the former corresponds to the pairs q and v , which have a higher correlation with each other. The parameter θ_1 relates the pair q and v with the duration d . In this context, maintaining duration as independent is of utmost importance, as it allows for more accurate control over this variable. It can be treated as a constant with the properties of the asymmetric copula. The next step is to define the appropriate duration for estimating hydrographs. The values of the parameters θ_1 and θ_2 , as well as the results of the Kolmogorov-Smirnov (KS) and Anderson-Darling (AD) tests, are presented in Table 6.

Figure 6 presents a graphical comparison between the OCF, $C(u, v, d)$, and $C(C(u, v), d)$. It is observed that the copulas effectively represent the observed data, with some spikes highlighting events whose characteristics deviate from the typical behavior observed in other years.

Table 6 | θ parameters obtained with the maximum pseudo-likelihood method for the copulas analyzed and their respective KS and AD values for the goodness-of-fit tests

Copula	θ_1	θ_2	KS	AD
Clayton symmetric	0.6894	na	0.1047	0.2168
Clayton asymmetric	0.2959	2.1230	0.1044	0.2160
Gumbel-Hougaard symmetric	1.1514	na	0.1147	0.2922
Gumbel-Hougaard asymmetric	1.1551	1.3122	0.1046	0.2686

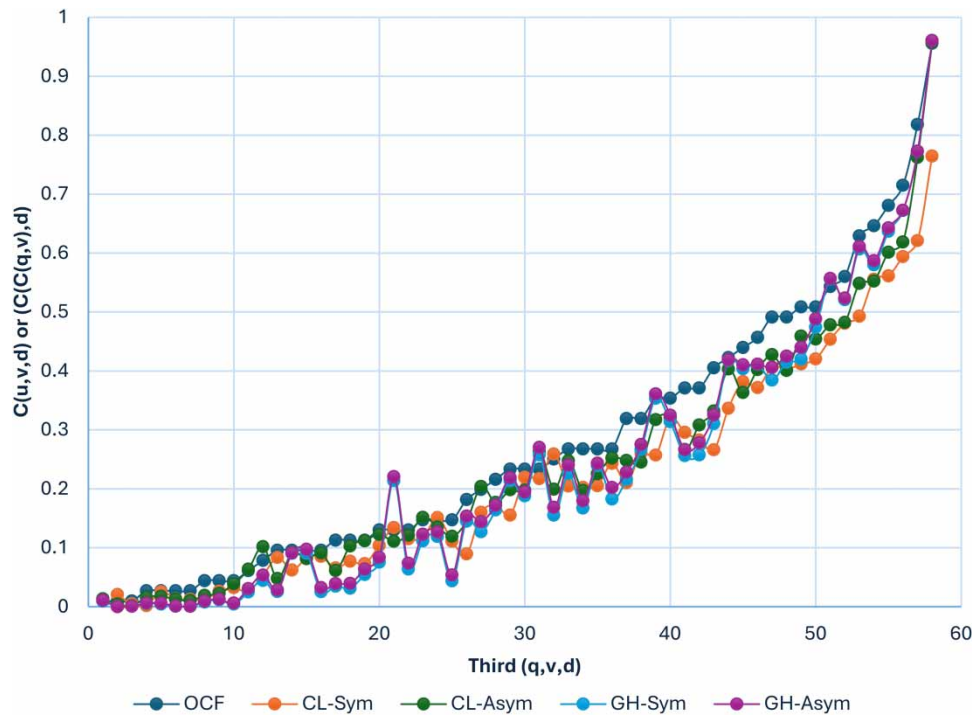


Figure 6 | OCF obtained vs. the values of $C(q, v, d)$ and $C(C(q, v), d)$ calculated with the Clayton and Gumbel-Hougaard symmetric and asymmetric copulas.

Trivariate extrapolation

As noted in previous sections, the duration variable has a much smaller range compared with the other variables, complicating the extrapolation process. Ensuring that duration remains within realistic values is crucial for maintaining the model's validity.

To perform the extrapolation, the return period function OR and Kendall's τ were used, with return periods set to 100 and 200 years. This strategic choice allows for a robust evaluation of the estimation over the long term. With 58 years of recorded data, the 100-year return period enables testing the reliability and predictive power of the copula models in accurately capturing extreme events and their dependencies.

The focus on maintaining realistic values for duration is crucial because any deviation can lead to significant inaccuracies in the model's predictions. By carefully selecting the return period and ensuring that the duration variable does not extrapolate to unrealistic values, we aim to enhance the precision and robustness of our model. This approach ensures that the extrapolated values are consistent with the observed historical data and provides confidence in the model's ability to predict future events under similar conditions.

Given the inherent complexity of trivariate extrapolation, the process can be simplified by treating one of the variables as a constant. By doing so, the joint probability will be satisfied when the remaining two variables align with the specified probability.

Duration is the ideal candidate for this constant treatment because it has been thoroughly analyzed in terms of its marginal probability and exhibits a relatively smaller range compared with the other variables. This prior analysis provides a high degree of control and reliability over the duration variable. Consequently, we will fix the duration variable at constant values ranging between 12 and 18 days. These durations have not been arbitrarily chosen; on the contrary, given the model with the copula, they are those for which the joint probability associated with volume and discharge is satisfied. They correspond to the Weibull marginal function and represent an appropriate value with respect to the observed events. In other words, the range of duration established based on the copula represents real values for the behavior of the river under study.

Holding duration constant simplifies the extrapolation process by focusing on the interaction between the remaining two variables, reducing computational complexity and improving result clarity. This approach ensures that duration stays within realistic bounds, preventing unrealistic extrapolation.

To address this, isolines for joint return periods OR and Kendall's tau have been generated. These isolines account for the fact that various combinations of flow (q) and volume (v) can achieve the joint probability for the specified return period, with fixed durations.

By plotting these isolines, we can visualize the range of combinations that satisfy the joint return period criteria, with durations set between 12 and 15 days for Kendall's return periods and 15–18 days for OR return periods. As discussed in Section four, both return periods are derived from the same expression but use different probability values. Figure 7 visually represents the behavior of the expression for $Tr = 1/1 - C$, where the value of (t) is embedded within the function taking the value of $C(C(u, v), d)$. These values enable the extraction of both return periods.

To aid understanding and use the values for the desired return period (100 and 200 years in this study), the values of u and v that meet the specified probability were extracted and plotted on a $q - v$ plane. This approach provides a clear view of how different pairs of flow and volume interact under the fixed duration constraint to achieve the desired return period and its associated probability.

The trivariate isolines, shown in Figures 8 and 9, illustrate these relationships, helping to understand the dependency structure and the range of plausible scenarios that meet the joint probability requirements. This visualization is crucial for

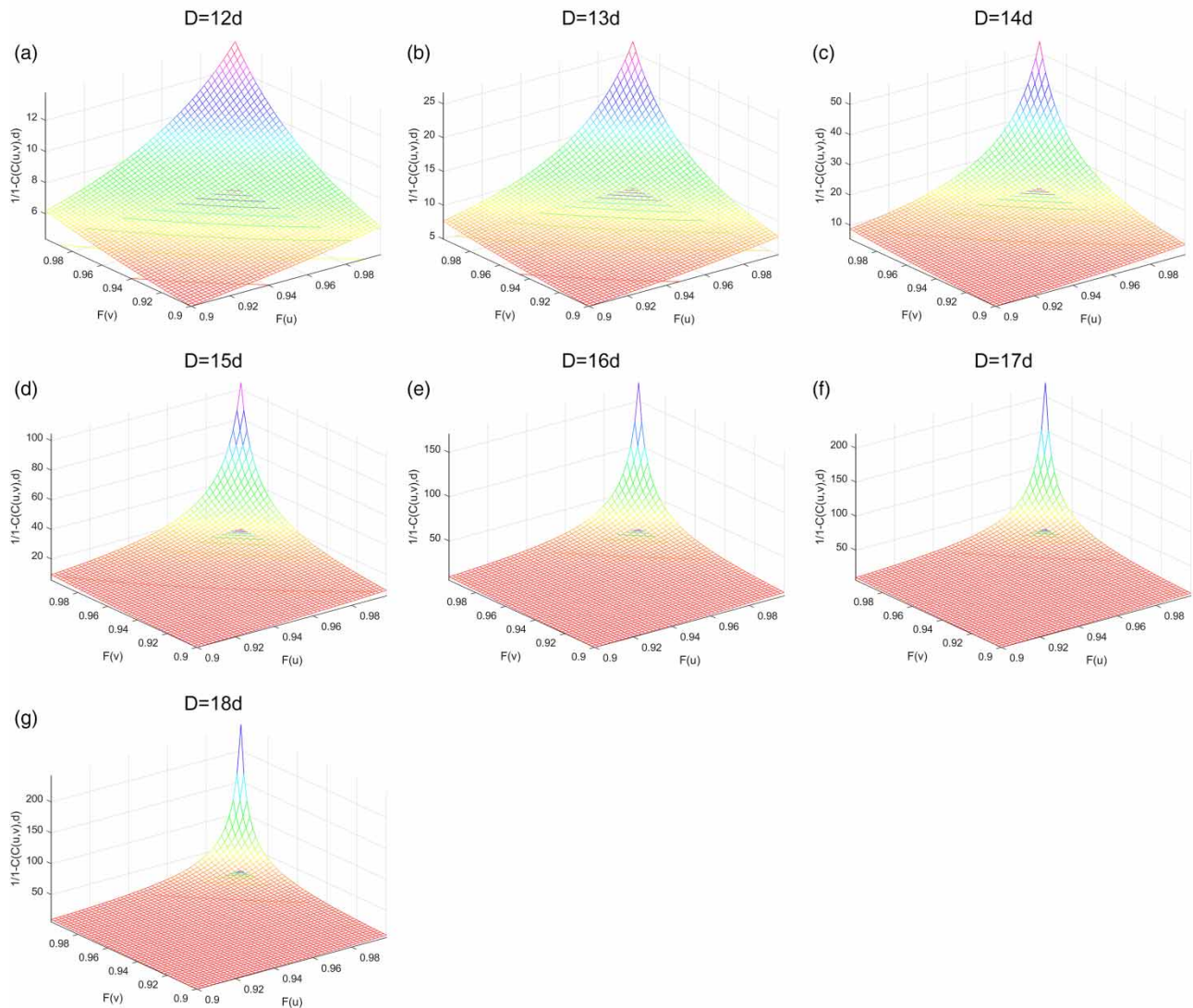


Figure 7 | Joint return periods for the selected durations using the asymmetric Clayton copula.

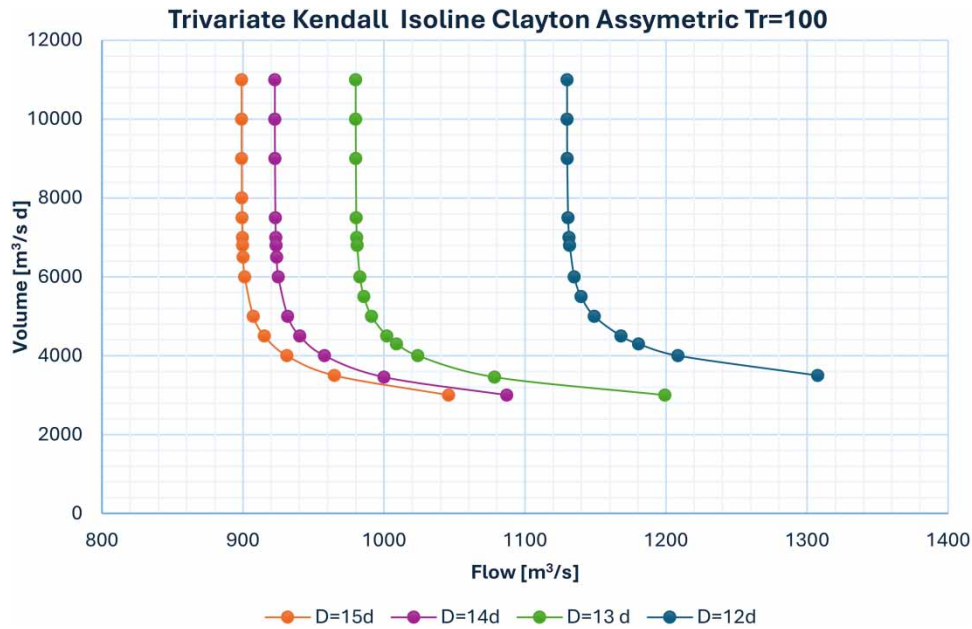


Figure 8 | Isolines calculated with Tr_{ken} for 100 years for durations equal to 12, 13, 14, and 15 days.

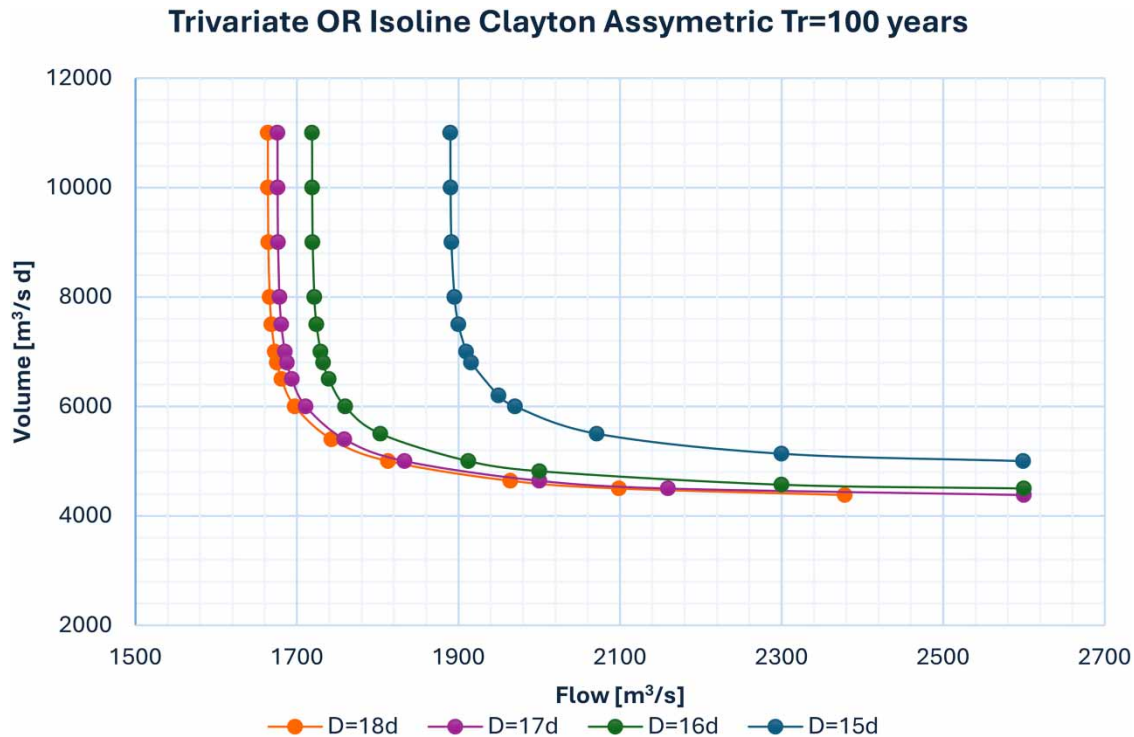


Figure 9 | Isolines calculated with Tr_{OR} for 100 years for durations equal to 15, 16, 17, and 18 days.

communicating the results of the trivariate extrapolation and aids in decision-making and risk assessment. Figures 10 and 11 show similar isolines for the 200-year joint return period.

It is evident that the isolines provide different values, primarily in terms of duration. The copula cannot model events with durations less than 15 days for the OR case, as it is constrained by the marginal probability and its connection to the other two

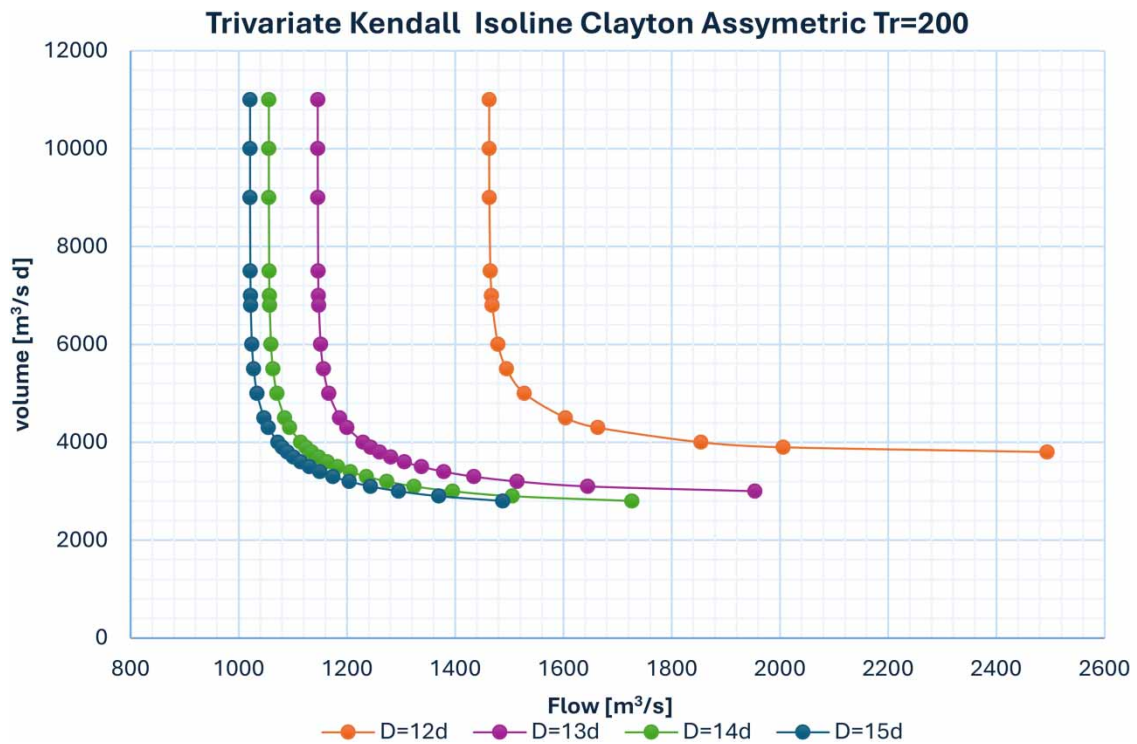


Figure 10 | Isolines calculated with Tr_{Ken} for 200 years for durations equal to 12, 13, 14, and 15 days.

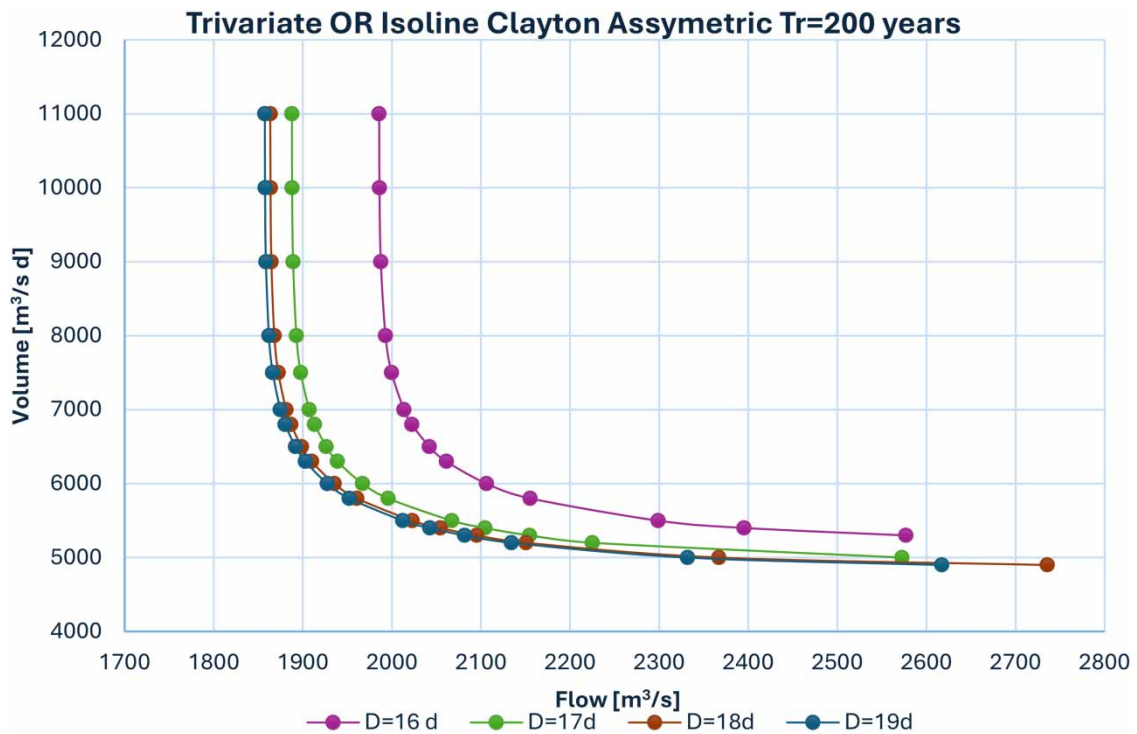


Figure 11 | Isolines calculated with Tr_{OR} for 200 years for durations equal to 16, 17, 18, and 19 days.

variables through the association parameter. Consequently, events are modeled with longer durations and higher flows compared with the isolines estimated with Kendall's return period.

The estimates using Kendall's return period are more consistent with observed values, resulting in a probabilistic representation that better reflects the natural behavior of events. This approach offers a more accurate and realistic depiction of the interactions between flow, volume, and duration, leading to more reliable predictions and assessments.

Shaping the hydrograph

Based on the isolines generated for both joint return period definitions, representative combinations of flow, volume, and duration were selected. Specifically, the points with the maximum curvature along each isoline were chosen to provide optimal contrast for analysis. The selected values are summarized in Table 7.

The shape of the design hydrograph is critical for hydraulic infrastructure design, operational planning, and flood risk assessment. The time to peak is particularly relevant, as it determines when the maximum inflow reaches the reservoir. A shorter time to peak implies higher instantaneous storage demand, while a longer time may lead to sustained inflows that gradually fill the reservoir. This dynamic is directly linked to the total volume of runoff entering the system.

To construct hydrographs that exactly satisfy the selected extrapolated values of peak flow, volume, and duration, the system of equations previously described in Section two was employed. This method allows the application of a predefined hydrograph shape while ensuring mathematical consistency with the probabilistic values derived from the copula model. The shape implemented in this study is based on a Pert-like function, adapted by the Engineering Institute of the National Autonomous University of Mexico (UNAM) to allow for flexible fitting of the three variables involved.

Figures 12–15 present the resulting hydrographs for both OR-based and Kendall-based return periods, at 100- and 200-year recurrence intervals. As anticipated, hydrographs associated with Kendall's return period exhibit lower peak flow, volume, and duration values than those derived from the OR-based return period, reflecting the more conservative nature of the Kendall framework.

For both cases, as the duration increases, the flow and volume decrease. This is due to the direct impact of the marginal probability of duration on the joint probability estimated with the Clayton copula. Specifically, events with 100-year return periods have been estimated, where the values of the two remaining variables increase with shorter durations. This may depend on the circumstances under which adverse outcomes are observed, as higher flow and volume values occur over shorter time periods. Conversely, it may be disadvantageous for lower flow and volume values to persist over longer durations. Therefore, when estimating these trivariate hydrographs, their results must be considered individually for each case.

Additionally, the duration directly influences the peak time of the hydrograph due to the selected shape. The Pert shape allows for adjustment to flow and volume values according to the established duration. The peak time for this shape is a function of the total duration, so each hydrograph 'shifts', further reducing its volume and discharge when considering the increase in duration as the main variable.

As is expressed by Serinaldi (2015) 'the primary OR return period, and secondary Kendall's return period describe different situations, and their use depends on which one better describes the design requirements and mechanisms of failure. The Tr_{OR}

Table 7 | Thirds Q , V , D for joint return period of 100 and 200 years selected from the isolines Tr_{Ken} and Tr_{OR}

Tr_{Ken}			Tr_{OR}		
D (d)	Q (m ³ /s)	V (m ³ /s)	D (d)	Q (m ³ /s)	V (m ³ /s)
15	931.10	4,000	15	1,949.43	6,200
14	940.36	4,500	16	1,803.27	5,500
13	996.71	4,700	17	1,758.67	5,400
12	1,149.13	5,000	18	1,742.83	5,400
15	1,080.43	3,900	19	1,902.92	6,300
14	1,114.52	4,000	18	1,910.02	6,300
13	1,186.48	4,500	17	1,926.35	6,500
12	1,527.78	5,000	16	2,042.01	6,500

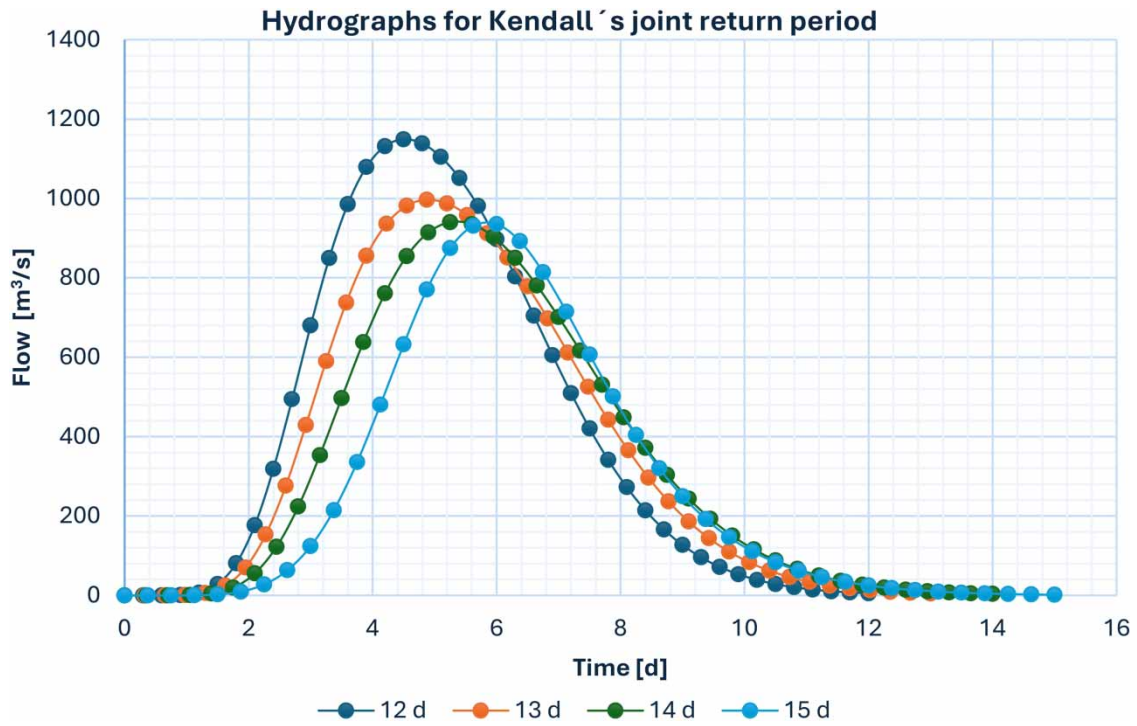


Figure 12 | Trivariate hydrographs estimated for Tr_{Ken} for a 100-year joint return period.

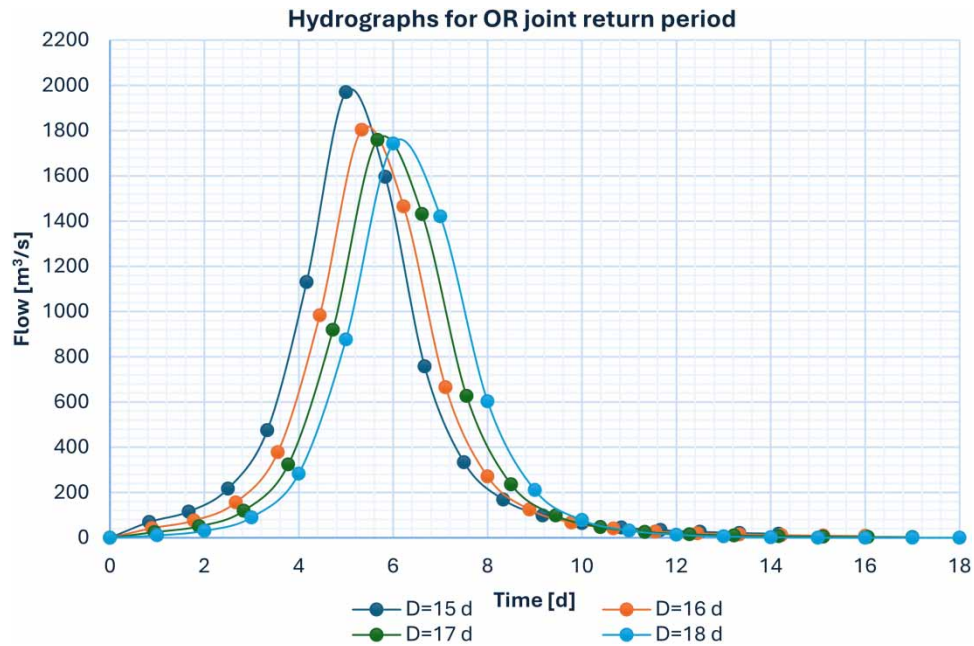


Figure 13 | Trivariate hydrographs estimated for Tr_{OR} for a 100-year joint return period.

defines the probabilities associated with critical regions defined by a unique pair of values (u, v) (in this case, the third variable is constant), Tr_{Ken} measures the probability associated with critical regions defined by an infinite set of points lying on a t -level curve. The design criterion focuses on (u, v) in the second on t value. In design terms, the primary return period the pair (u, v) guarantees a prescribed joint probability of exceedance, provided that a failure occurs when both values, u and v

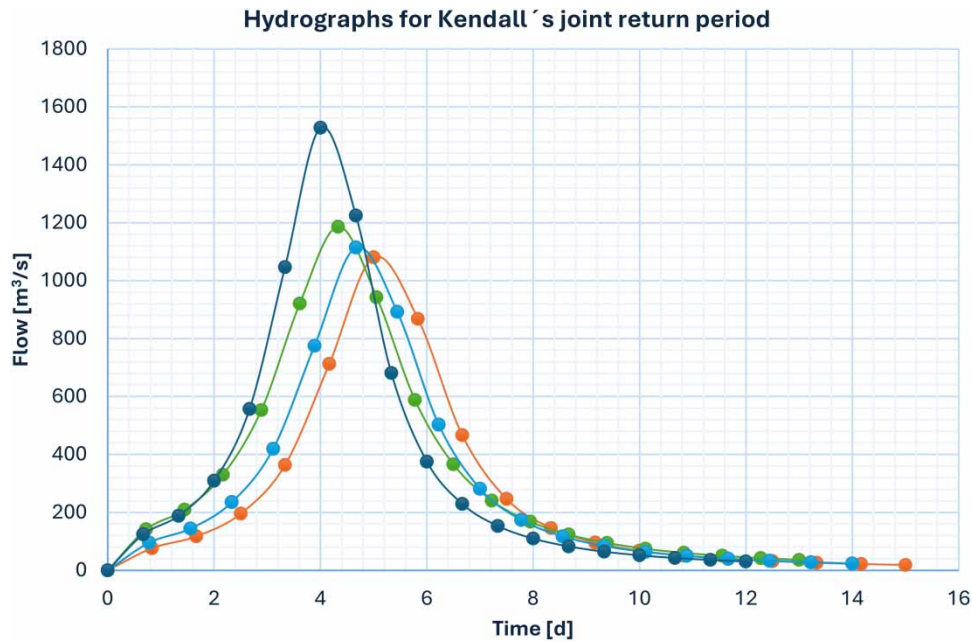


Figure 14 | Trivariate hydrographs estimated for Tr_{Ken} for a 200-year joint return period.

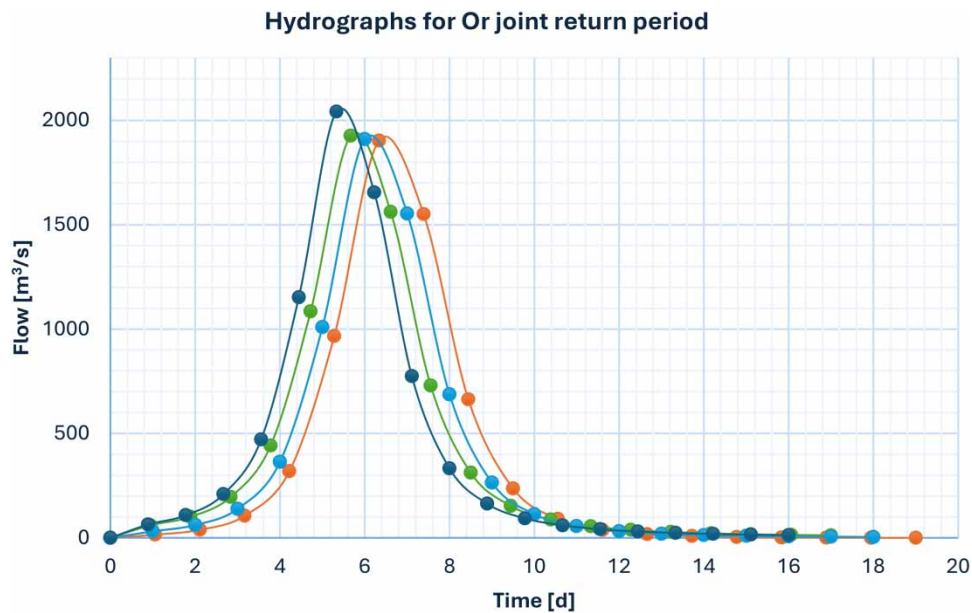


Figure 15 | Trivariate hydrographs estimated for Tr_{OR} for a 200-year joint return period.

are exceeded. The secondary return period describes a system for a set of bivariate events that is characterized by the same joint probability exceedance'. According to this, the use of the hydrographs with Tr_{OR} or Tr_{Ken} will yield different results because their considerations are different, a design with the OR results will require that both variables q, v are exceeded, contemplating that the probability of exceedance of the third variable, for the given duration, will also be exceeded. The use of Kendall hydrographs will contemplate that a set of three variables could be surpassed. The choice between these methods depends on the specific objective of the analysis.

Flood routing analysis for the La Angostura reservoir

Using the modified Pert hydrographs for each duration, flood routing through the La Angostura reservoir estimates the maximum elevation reached and the corresponding stored volume. Figure 16 illustrates that hydrographs derived from the OR return period reach higher elevations, a result of their derivation method. Comparisons between Kendall and OR hydrographs show that shorter durations lead to higher reservoir elevations due to larger inflow discharge and volume occurring over a shorter time, prompting a quicker reservoir response and increased water levels.

The impact of duration changes is more pronounced with Kendall hydrographs, which show greater variability in elevation compared with the more stable OR hydrographs (Figure 17). Therefore, for design or review purposes, the shortest durations in both cases should be prioritized.

In Tables 8 and 9, the inflow hydrographs to the Angostura Dam reservoir are presented for each duration defined for the Kendall and OR return periods, including peak discharge, inflow volume, and peak time. Additionally, the outflow hydrographs are provided, detailing the maximum elevation reached in the reservoir and the maximum stored volume. These data provide a comprehensive understanding of the reservoir's behavior during the flood event, enabling a thorough assessment of inflow and outflow rates as well as the reservoir's storage capacity over the study period.

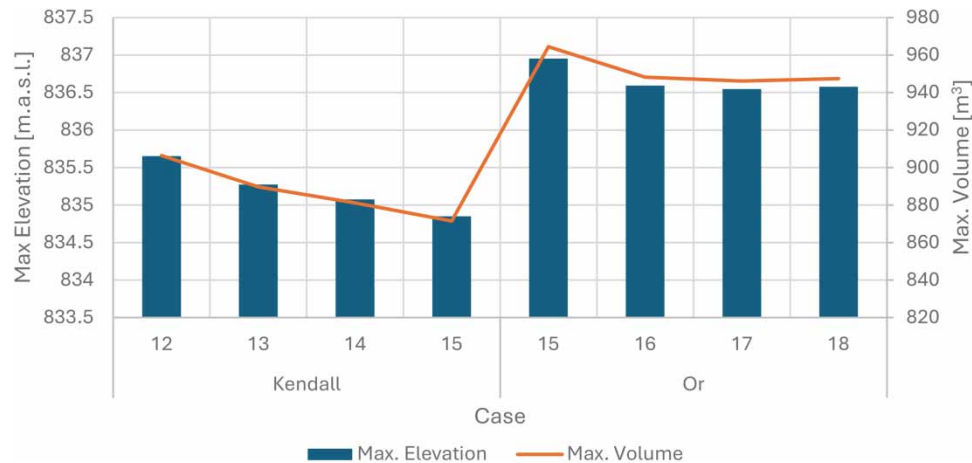


Figure 16 | Maximum elevation and volume reached in the La Angostura reservoir starting from the ordinary maximum water level after flood routing of Kendall and OR for 100 years hydrographs for durations of 12–18 days.

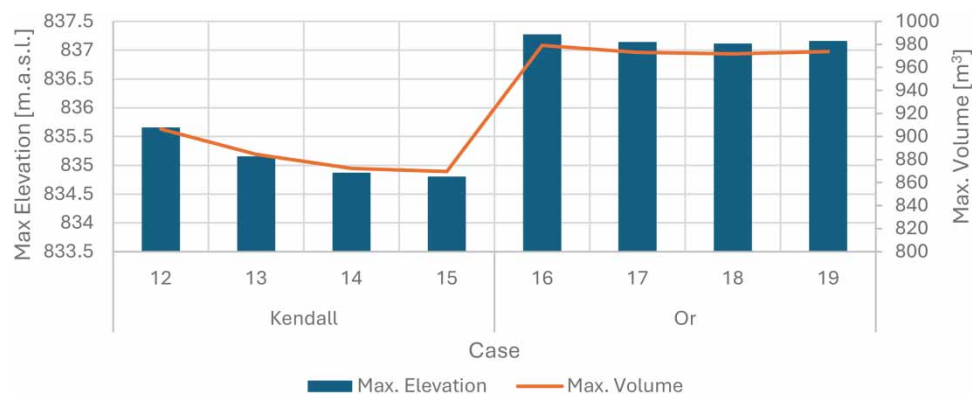


Figure 17 | Maximum elevation and volume reached in the La Angostura reservoir starting from the ordinary maximum water level after flood routing of Kendall and OR for 200 years hydrographs for durations of 12–19 days.

Table 8 | Flood routing results: inflow and outflow hydrographs, maximum reservoir elevations, and volumes for 100-year joint return period

Joint return period		Inflow hydrograph duration (d)	Inflow peak discharge (m ³ /s)	Inflow volume (hm ³)	Inflow peak time (d)	Outflow peak discharge (m ³ /s)	Outflow peak time (d)	Outflow duration (d)	Outflow volume (hm ³)	Max. elevation (masl)	Max. stored volume (hm ³)
Tr = 100 years	Ken	12	1,149.130	437.20	4.5	721.45	6.542	20	428.355	835.65	906.41
		13	996.710	410.64	4.875	640.45	7.042	20	401.502	835.27	889.78
		14	940.360	392.79	5.25	598.68	7.417	20	383.380	835.08	881.11
		15	935.320	348.71	6	553.76	7.708	20	339.436	834.85	871.48
	OR	15	1,969.900	521.78	5	1,023.42	6.417	25	511.926	836.95	964.48
		16	1,803.270	477.89	5.33	935.89	6.792	25	468.261	836.59	948.22
		17	1,758.670	469.32	5.66	925.1	7.208	25	459.719	836.55	946.14
		18	1,742.830	470.14	6	932.12	7.583	25	460.467	836.58	947.49

Table 9 | Flood routing results: inflow and outflow hydrographs, maximum reservoir elevations, and volumes for 200-year joint return period

Joint return period		Inflow hydrograph duration (d)	Inflow peak discharge (m ³ /s)	Inflow volume (hm ³)	Inflow peak time (d)	Outflow peak discharge (m ³ /s)	Outflow peak time (d)	Outflow duration (d)	Outflow volume (hm ³)	Max. elevation (masl)	Max. stored volume (hm ³)
Tr = 200 years	Ken	12	1,527.785	432.00	4	722.31	5.292	20	397.517	835.66	906.58
		13	1,186.486	388.80	4.33	615.46	5.708	20	385.702	835.16	884.61
		14	1,114.522	345.60	4.66	557.89	6.125	20	339.839	834.87	872.36
		15	1,080.431	336.96	5	544.76	6.5	20	329.726	834.81	869.55
	OR	16	2,042.019	561.60	5.33	1,103.14	6.792	25	556.283	837.27	979.15
		17	1,926.355	561.60	5.66	1,069.54	7.167	25	555.083	837.14	973.02
		18	1,910.027	544.32	6	1,063.17	7.583	25	537.316	837.12	971.85
		19	1,902.924	544.32	6.33	1,074.69	7.958	25	537.499	837.16	973.85

DISCUSSION

The results of this study highlight the importance of explicitly incorporating duration as a variable in trivariate hydrological frequency analysis, alongside peak flow and runoff volume. Unlike conventional approaches that treat duration as fixed or visually inferred, this study models it statistically, enhancing the consistency and representativeness of design hydrographs while overcoming the limitations of bivariate methods.

Compared with prior studies, this work refines trivariate dependency modeling. Zhang & Singh (2007) used a symmetric Gumbel-Hougaard copula but did not address how to extrapolate or prioritize duration, possibly limiting robustness. Grimaldi & Serinaldi (2006) introduced asymmetric copulas, showing that symmetric models may misrepresent dependence when one variable is weakly correlated. In this case, duration shows the weakest correlation, supporting the use of asymmetric copulas – specifically Clayton – to better capture the dependence structure.

This study advances beyond visual event selection methods (e.g., Ganguli & Reddy 2013; Latif & Mustafa 2020) by proposing a hybrid method that combines annual maxima and threshold exceedance, reduces subjectivity, prioritizes duration, and preserves the joint structure, producing more coherent trivariate datasets.

Salvadori *et al.* (2011) proposed simplified models treating duration as conditionally independent $C_{QVD}(u, v, d) = C_{QV}(u, v) \cdot d$, but our results reject this assumption. Even with weak dependence, duration structurally interacts with peak flow and volume, justifying a complete trivariate copula.

The choice of marginal distributions is also crucial. Since duration spans a narrow range, heavy-tailed functions (e.g., Gumbel-Gumbel) may yield unrealistic extrapolations. This study uses the Weibull distribution for duration due to its tail stability, which prevents overestimation and supports more accurate joint extrapolations.

Both symmetric and asymmetric forms of Clayton and Gumbel-Hougaard copulas were tested. The asymmetric Clayton copula gave the best fit, particularly for the highly dependent Q - V pair. However, this is case-specific; other studies (e.g., Ganguli & Reddy 2013; Bezak *et al.* 2014) report better results with Student's t and Gumbel copulas, emphasizing the need for case-specific selection.

The study compares isolines from OR-based and Kendall return periods. OR-based copulas failed to model events shorter than 15 days due to constraints in marginal duration probability, producing longer and more voluminous hydrographs. Kendall return periods captured shorter durations more accurately and reflected observed behavior better. These findings stress the importance of choosing the return period method based on design objectives and failure mechanisms.

Flood routing with Pert hydrographs showed that Kendall-derived hydrographs produced greater variability in reservoir peak elevations than the more stable OR-based versions. Shorter durations yielded higher peaks, underscoring their relevance in dam safety assessments and justifying their prioritization in design and review processes.

Overall, the study introduces a robust methodology that positions duration as a core variable in trivariate frequency analysis. It improves on previous approaches by refining event selection, adapting marginal fitting, using asymmetric copulas to capture realistic dependence, and offering a flexible framework for hydrograph shaping. The resulting hydrographs are statistically consistent and hydraulically interpretable, aiding flood risk evaluation, infrastructure design, and reservoir management.

However, limitations persist. Multi-peaked events may inflate volumes, marginal fitting may be challenged by sparse data, and fixing duration during extrapolation introduces subjectivity that must be addressed transparently.

A flexible algebraic shaping model was proposed to generate design hydrographs. It allows tailoring hydrograph shapes while satisfying estimated Q , V , and D values, avoiding fixed templates and enabling realistic routing. Though less probable, shorter durations lead to higher peaks and volumes, underscoring their critical role in design. Longer durations yield attenuated but less critical hydrographs. Thus, duration is vital in dam safety, storage planning, and emergency protocols.

This study delivers a replicable methodology for trivariate hydrological frequency analysis, centered on duration. It advances prior models by addressing key gaps in variable selection, dependence modeling, and hydrograph generation – contributing to more informed and reliable hydraulic design under uncertainty.

CONCLUSIONS

This study confirms the critical role of event duration in trivariate hydrological frequency analysis and design hydrograph estimation. By treating duration as a statistically modeled variable – rather than an assumed or visually determined input – the methodology enhances the internal consistency and predictive reliability of multivariate flood modeling. The hybrid data selection method prioritizes duration while preserving the natural correlation structure among peak flow, runoff volume, and duration, improving both data coherence and estimation accuracy.

The analysis shows that shorter-duration events, though less frequent, result in higher peaks and volumes, and thus pose a greater challenge for infrastructure design and flood routing. In contrast, longer-duration events produce attenuated hydrographs, which are easier to manage but less critical in terms of reservoir stress and operational risk. This behavior underscores the importance of accurately modeling duration, particularly when defining synthetic design scenarios.

The use of asymmetric Clayton copulas was found to be the most appropriate for the case study, especially for the $Q - V$ pair, which exhibited the strongest dependence. However, this finding is case-dependent and should not be generalized, as other studies have found different copulas to be more suitable depending on hydrological conditions. The careful selection of marginal distributions, especially for the duration variable, was also essential to avoid unrealistic extrapolations. The Weibull distribution provided stable tail behavior and supported credible return period estimations.

Joint return period analysis further revealed that Kendall-based isolines better represented observed events compared with those derived from OR-based formulations, particularly in capturing short-duration scenarios. Flood routing simulations using the estimated hydrographs showed that Kendall-derived hydrographs led to greater variability in peak reservoir elevation, reinforcing the need to prioritize short-duration events in design and emergency planning.

In conclusion, this work contributes a unified and flexible methodology for trivariate hydrological frequency analysis, emphasizing the role of duration, improving dependency modeling through asymmetric copulas, and offering a generalizable framework for synthetic hydrograph generation. The approach is well-suited for applications in flood risk assessment, dam operation, and hydraulic infrastructure planning under uncertainty.

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AUTHOR CONTRIBUTIONS

The document was jointly conceived, developed, and ideated by both authors. E.O.C.M. was responsible for the meticulous data selection and methodology. Ó.A.F.M. conducted comprehensive joint probability analyses. Additionally, E.O.C.M. and Ó.A.F.M. collaborated on the intricate calculations pertaining to trivariate characteristics. The manuscript was meticulously drafted by E.O.C.M., ensuring clarity and coherence throughout. Both authors diligently reviewed and provided their approval of the final manuscript, ensuring its accuracy and integrity.

STATEMENT OF AI TOOL USAGE

This paper has undergone review with the assistance of an artificial intelligence tool, specifically employed to enhance translation, grammar, and punctuation in writing. The content of the text has been developed, reviewed by the authors, and is the responsibility of them.

DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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